PID Control Design for Power Factor Correction in an AC–DC Boost Converter

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***Abstract*—Switched-mode front ends fed directly from a diode bridge draw current in short, peaky pulses that distort the line waveform and pull the displacement angle away from zero, so the power factor presented to the utility rarely exceeds 0.6–0.7 unless some form of active shaping is added. This work documents the design of a dual-loop, PID-controlled boost-type power factor correction stage that forces the input current to follow the rectified line voltage while holding the 400 V output bus steady against load and supply disturbances. An outer voltage loop sets the amplitude of a sinusoidal current template derived from the sensed AC source, and an inner current loop forces the inductor current onto that template at every switching instant; both loops use proportional–integral–derivative compensators with the derivative path disabled, which is common practice when the noise floor of the sensed current makes derivative action counter-productive. The complete plant, comprising the boost inductor, output capacitor and load, together with both control loops, was built as an averaged continuous-time model in MATLAB/Simulink and exercised at 230 V, 50 Hz input with an 800 W resistive load switching at 100 kHz. The closed-loop system settles to a mean output of 400.25 V (0.062% regulation error) with 14.32 V peak-to-peak ripple, an input power factor of 0.9412 and a current total harmonic distortion of 20.38%, the last of which falls short of the IEC 61000-3-2 Class A limit and is examined in detail. The gap between the achieved and the standard-compliant THD is used to motivate a comparison against more elaborate compensation schemes reported in the recent literature, several of which trade implementation simplicity for a tighter harmonic budget.**

***Index Terms*—Boost converter, harmonic distortion, PID control, power factor correction, total harmonic distortion, voltage regulation**

Nomenclature

$v_{in}(t)$	instantaneous line voltage
V_{ac}	rms value of the input voltage
V_{out}, V_{ref}	output bus voltage, output set-point (400 V)
L, C, R	boost inductor, output capacitor, load resistance
$i_L(t)$	instantaneous inductor (input) current
$i_{ref}(t)$	current-loop reference template
$d(t)$	switching duty ratio, $0 \leq d \leq 1$
f_{sw}, f_{line}	switching frequency, line frequency
K_p, K_i, K_d	proportional, integral, derivative gains
N	derivative filter coefficient
$e_v(t), e_i(t)$	voltage-loop and current-loop error signals
P, Q, S	real, reactive and apparent power
I_h	rms magnitude of the h^{th} current harmonic
ω_n, ζ	natural frequency and damping ratio

Acronyms

PFC	Power Factor Correction
PID	Proportional–Integral–Derivative
PI	Proportional–Integral
THD	Total Harmonic Distortion
DPF	Displacement Power Factor
CCM	Continuous Conduction Mode
DCM	Discontinuous Conduction Mode
ACMC	Average Current Mode Control
PSO	Particle Swarm Optimization
GWO	Grey Wolf Optimization
FOPID	Fractional-Order PID
ANN	Artificial Neural Network
FLC	Fuzzy Logic Controller
SMC	Sliding Mode Control
MPC	Model Predictive Control
EV	Electric Vehicle

I. Introduction

A switching converter fed by an uncontrolled diode bridge and a large filter capacitor is, electrically, a bad neighbour on the supply network. It only draws current near the peak of the line voltage, when the capacitor needs topping up, so the current waveform collapses into narrow pulses several times taller than the equivalent sinusoid would need to be. The displacement between voltage and the fundamental of that pulsed current is usually small, but the harmonic content is not, and the two effects combine to push the power factor down toward 0.5–0.65 for a typical capacitor-input rectifier. Utilities care about this because the reactive and harmonic components occupy distribution capacity without doing any useful work, and regulators have written that concern into standards such as IEC 61000-3-2 and IEEE 519, which cap the harmonic current a piece of equipment may inject back into the mains.

Active power factor correction addresses the problem by inserting a controlled converter, almost always a boost stage, between the rectifier and the bulk capacitor. The boost topology is the default choice for this job for reasons that have little to do with elegance and a lot to do with practicality: it needs one switch, one diode and one inductor, its input current is continuous rather than pulsed, and its output voltage is by construction higher than the rectified peak, which keeps the control problem well posed across the whole AC cycle. Once the topology is fixed, the remaining design decision is how to control the duty cycle so that the inductor current tracks a scaled, rectified copy of the line voltage while the output capacitor voltage is held at a constant DC level under load and line disturbances.

Several families of controllers have been proposed for this dual objective, ranging from hysteresis and peak-current control through sliding-mode and predictive schemes to metaheuristically tuned fractional-order compensators. Classical PID control sits at the simpler end of that spectrum, and it remains the default starting point in undergraduate laboratories and in a large share of commercial PFC integrated circuits, not because it is the best performer on paper but because its structure is transparent, its tuning rules are well documented, and it can be implemented on hardware with no more computational muscle than an 8-bit microcontroller or a handful of operational amplifiers. The relevance of PID-based PFC has not faded as more exotic alternatives have appeared in the literature; instead, it has settled into a particular niche, namely cost-sensitive or pedagogically motivated designs where a few percent of extra harmonic margin is less important than predictable behaviour, easy commissioning and a controller a maintenance technician can actually reason about without a background in optimization theory. Compared with passive compensation (bulky line-frequency inductors or capacitor banks that only correct the displacement component and do nothing for harmonics), a PID-driven boost stage corrects both the displacement and distortion components simultaneously and adapts automatically to load changes, which passive networks cannot do.

This paper reports the design, the closed-loop equations and the simulated performance of such a dual-loop PID-controlled boost PFC stage, built and exercised entirely in MATLAB/Simulink using an averaged behavioural model of the power stage. Section II surveys recent work on PFC control, ranging from classical PI schemes through metaheuristically tuned and fractional-order variants. Section III states the control problem precisely. Sections IV through VI cover the system architecture, the governing equations and the PID design itself. Section VII describes the Simulink implementation, and Section VIII reports the simulated power factor, harmonic content, output regulation and transient response. Sections IX through XI compare the result against the literature, weigh its advantages and shortcomings, and lay out directions for future refinement, before Section XII concludes the paper.

II. Literature Review

The research base on power-factor-correcting converters is large enough that a single classification scheme rarely does it justice; the following discussion groups recent contributions by the kind of controller employed, moving from topology surveys through classical and metaheuristically tuned PID/PI schemes to fractional-order and intelligent variants, before closing with a comparison table.

Topology choice is the first decision a designer makes, and several recent surveys have organized the available options. Ortiz-Castrillón *et al.* [1] classified single-phase boost-derived PFC converters into bridge, semi-bridgeless and fully bridgeless families and tabulated their relative THD, efficiency and component count, concluding that the conventional bridge boost converter, despite carrying two extra rectifier diodes, remains attractive whenever simplicity and cost outweigh the last few tenths of a percentage point of efficiency. Chen, Davari and Wang [2] pushed the same comparison further for bridgeless variants specifically, deriving a unified framework that benchmarks common-mode noise, conduction loss and control complexity across topology families, and noted that bridgeless circuits trade lower conduction loss for control schemes that must separately handle the two AC half-cycles. Kim, Choi and Won [3] took a narrower view, proposing a modulated-carrier control scheme for a conventional boost PFC converter that removes the multiplier circuit found in classical average-current control, at the cost of a more involved carrier-generation block; their simulated and experimental results show power factor above 0.99 but the paper does not examine behaviour under fast load steps. Espitia's 2025 systematic review [4], built on a PRISMA-style search of six converter topologies, observed that the bulk of recently published PFC work still reports simulation-only validation and called for more comparative studies that quantify controller robustness rather than just steady-state harmonic figures, a gap the present paper partially addresses by reporting transient settling behaviour alongside the steady-state numbers.

Closer to the controller itself, Ortatepe and Karaarslan [5] compared a conventional PI controller against fuzzy self-tuning and sliding-mode control on a bridgeless PFC converter under unbalanced grid conditions, and found that the conventional PI loses ground specifically when the supply voltage is distorted or unbalanced, while the fuzzy self-tuning approach holds THD low across both loops at the price of added tuning effort; their work leaves open how a PID controller with feedforward compensation might narrow that gap without resorting to a rule-based fuzzy layer. Ibrahim [6] stayed with a purely PI-based design but compared an adaptive-gain PI against a fixed Ziegler–Nichols-tuned PI on a hysteresis-current-controlled boost PFC converter, reporting that the adaptive variant cut output overshoot from roughly 250% down to about 50% during a voltage step, a result that underlines how sensitive a fixed-gain controller's transient response is to the operating point it was tuned at. Jensen *et al.* [7] proposed a time-based control scheme that replaces the analog current-loop compensator with a digital on-time calculation, sidestepping PID tuning altogether for the inner loop; the approach simplifies the controller but ties the design tightly to a specific digital implementation, which limits its portability to other hardware. Molavi, Maghsoudi and Farzanehfard [8] pursued a different route again, using a quasi-resonant bridgeless topology to push input-current THD down through soft-switching rather than through more aggressive control, a reminder that some of the harmonic budget in any PFC design is set by the power stage itself before the controller ever gets a chance to act on it.

A second cluster of papers tunes the gains of a PID or PI controller using bio-inspired search rather than analytical or trial-and-error methods. Romero, Páez, Noriega and co-authors [9] applied a particle-swarm-based search directly to PID sintonization for a boost PFC converter, reporting improved power factor over a manually tuned baseline but without an experimental implementation. The same research group later extended the idea with a Multi-Particle Swarm Optimization algorithm in Guarnizo *et al.* [10], tuning all four parameters of a Simulink PID block (including the derivative filter coefficient N) for a 100 W boost PFC stage and validating the result on a physical prototype; their central finding is that offline metaheuristic tuning avoids the risk of destabilizing the converter during an online search, which is an important practical consideration whenever the plant being tuned is also the equipment being protected. Demirtas and Ahmad [11] compared a fixed-gain PI, a fuzzy-logic PI and a fractional-order PI for a 600 W boost PFC converter, all three tuned by particle swarm optimization, and found the fractional-order variant gave the best combination of power factor and harmonic performance among the three, at the cost of an extra non-integer-order parameter that complicates real-time implementation. Kandasamy and Prem Kumar [12] moved to grey wolf optimization for tuning a bridgeless AC–DC converter's compensator, reporting efficiency gains over a conventionally tuned baseline, though their comparison stops short of reporting a harmonic spectrum, which makes it difficult to judge the result against IEC limits directly.

A third group of recent papers replaces the integer-order PID structure altogether with a fractional-order or learning-based compensator. Lai and Wang [13] tuned a fractional-order PID for a CCM boost PFC converter using a grey wolf algorithm and reported lower THD than an integer-order PID baseline across a range of load conditions, attributing the improvement to the extra degree of freedom the fractional order provides in shaping the loop's phase margin. Liu, Liu and Wang [14] cascaded an Ant-Lion-optimized fractional-order PID outer voltage loop with a neural-network-based inner current loop, arguing that the neural network's adaptive capability handles operating-point variation better than a fixed-gain inner loop, though their paper acknowledges that training and validating the network adds considerable design overhead compared with a conventional PI current loop. Vijayakumar and Sudhakar [15] applied Golden Eagle optimization to a fractional-order PI controller for a SEPIC-based PFC stage aimed at electric-vehicle charging, reporting a notably low THD of 1.85% and citing fast settling and reduced inrush current as the main benefits, which places their result well inside the IEC Class A envelope and considerably ahead of the figure obtained with the fixed-gain dual-PID design examined in this paper. Sánchez *et al.* [16] examined a fractional-order PID approximation built from passive RC networks and op-amps for a buck–boost converter, demonstrating that fractional control is not necessarily tied to a microcontroller-based implementation, while Mollaee *et al.* [17], Saadat *et al.* [18], Ghamari *et al.* [19], Roosta *et al.* [20] and Khavari [21] collectively explored adaptive, backstepping, Lyapunov-based and type-2 fuzzy variants of fractional or self-tuning PID control on buck and buck–boost converters; although these are not PFC-specific, they illustrate how actively the broader DC–DC control literature is moving away from fixed-gain PID, a trend the PFC-specific papers above mirror.

A final set of papers looks at predictive and observer-based alternatives. González-Castaño *et al.* [22] proposed a DC-voltage-sensorless predictive controller for a versatile buck-boost PFC rectifier, removing one voltage sensor at the cost of a more involved prediction model, while Boddu and Ghosh [23] integrated a PFC converter with a stand-alone inverter for microgrid use, reporting that the PFC front end's regulation quality directly affects the inverter's output waveform quality downstream, a coupling that is easy to overlook when the two stages are designed in isolation. Thirupura Sundari and Umamaheswari [24] used a predictive Riccati controller for a modular three-phase PFC converter in a wind-energy application, reporting strong power-quality improvement but at a control complexity well beyond what a PID-only design would require. Tabbat *et al.* [25] analyzed a high-efficiency step-up converter with zero-voltage switching, underscoring once more that power-stage design choices and control design choices both contribute to the final THD figure and cannot be fully separated when comparing results across papers.

Read together, these contributions point to a consistent pattern: fixed-gain PID and PI controllers remain the baseline against which every metaheuristic, fractional-order or learning-based alternative is measured, and that baseline typically clears unity-adjacent power factor without much difficulty while struggling to push THD below the high single digits unless it is helped along by feedforward compensation, a faster current

TABLE I
SUMMARY OF REVIEWED LITERATURE ON PFC CONTROL

Author(s)	**Year**	**Methodology**	**Main Finding**	**Research Gap**
Ortiz-Castrillón *et al.* [1]	2021	Topology survey of bridge/semi-bridgeless/bridgeless boost PFC	Bridge boost remains favoured for low-cost, low-complexity designs	Limited discussion of control-loop performance across topologies
Chen *et al.* [2]	2020	Comparative benchmarking of bridgeless topology derivatives	Quantified conduction-loss/efficiency trade-offs	Control design treated only briefly
Kim *et al.* [3]	2018	Modulated-carrier control, no analog multiplier	PF $>$ 0.99 with simplified hardware	Transient/load-step behaviour not reported
Espitia [4]	2025	PRISMA-based review of six PFC topologies	Most studies are simulation-only	Calls for robustness-focused comparative studies
Ortatepe and Karaarslan [5]	2020	PI vs. fuzzy self-tuning vs. SMC on bridgeless PFC	PI degrades under unbalanced/distorted grid	No feedforward augmentation explored for PI
Ibrahim [6]	2023	Adaptive PI vs. Ziegler–Nichols PI, hysteresis current control	Overshoot cut from $\sim$250% to $\sim$50%	Fixed-gain PI highly sensitive to operating point
Jensen *et al.* [7]	2019	Digital time-based control, no analog multiplier	Simplified current-loop hardware	Tied to specific digital platform
Molavi *et al.* [8]	2021	Quasi-resonant bridgeless soft-switching	Reduced THD via power-stage design	Control-loop contribution to THD not isolated
Romero *et al.* [9]	2021	PSO-tuned PID, boost PFC	Improved PF over manual tuning	No hardware validation
Guarnizo *et al.* [10]	2023	MPSO-tuned 4-parameter PID, 100 W prototype	Avoids instability risk of online tuning	Limited to single power/load point
Demirtas and Ahmad [11]	2023	PSO-tuned PI, fuzzy-PI, fractional PI	Fractional PI best PF/THD trade-off	Added fractional-order parameter complicates real-time use
Kandasamy and Prem Kumar [12]	2022	GWO-tuned bridgeless converter	Improved efficiency	Harmonic spectrum vs. IEC limit not reported
Lai and Wang [13]	2024	GWO-tuned fractional-order PID, CCM boost	Lower THD than integer-order PID	Computational cost of FOPID not quantified
Liu *et al.* [14]	2024	ALO-tuned FOPID outer loop + ANN inner loop	Adaptive inner loop tracks reference accurately	Network training overhead not discussed
Vijayakumar and Sudhakar [15]	2024	GEO-tuned fractional PI, SEPIC PFC	THD as low as 1.85%	Implementation complexity of GEO not addressed
González-Castaño *et al.* [22]	2021	Sensorless predictive control, buck-boost PFC	Removes one voltage sensor	Prediction model sensitive to parameter drift
Boddu and Ghosh [23]	2023	PFC + stand-alone inverter, microgrid	PFC quality affects downstream inverter output	Two stages rarely co-designed
Thirupura Sundari and Umamaheswari [24]	2025	Predictive Riccati control, 3-phase modular PFC, WECS	Strong power-quality improvement	Control complexity well beyond PID

loop, or a better power stage. Table I summarizes the methodology, principal finding and the gap left open by each of the PFC-specific studies discussed above.

III. Problem Statement

An uncompensated diode-bridge front end with a capacitive output filter draws current only while the bridge is forward biased, which in practice means a few milliseconds around each line-voltage peak. The resulting current pulses are rich in odd harmonics and carry a large in-rush component at every cycle, pushing the true power factor down to roughly 0.55–0.65 and the current THD well above 100% for a typical capacitor-input rectifier of this power class, even though the displacement angle between the voltage and the current pulse train stays close to zero. None of that current pulse shape is acceptable for equipment subject to IEC 61000-3-2, and the reactive burden it places on upstream transformers and wiring is real even where no formal standard applies.

The control problem this paper addresses is therefore twofold and the two halves pull against each other. First, the inductor current must be shaped into a sinusoid in phase with the line voltage across the entire conduction cycle, including the low-voltage region near each zero crossing where the available volt-seconds for current shaping shrink to almost nothing. Second, the output capacitor voltage must be held at a constant 400 V DC despite the inherent twice-line-frequency ripple that any single-phase PFC stage produces on its output, and despite step changes in load. A voltage loop fast enough to reject that ripple perfectly would inject a 100 Hz component back into the current-loop reference, which would itself become a harmonic source; a voltage loop slow enough to avoid that interaction responds sluggishly to genuine load steps. Resolving

TABLE II
DESIGN SPECIFICATIONS

Parameter	Target value
Input voltage, V_{ac}	230 V rms, 50 Hz
Output voltage, V_{ref}	400 V DC
Rated load	800 W (200 Ω at 400 V)
Switching frequency, f_{sw}	100 kHz
Output capacitor, C	470 μF
Target power factor	$\geq$ 0.95
Target current THD	$\leq$ 8% (IEC 61000-3-2, Class A)
Output regulation	$\leq$ 1% steady-state error

this tension is the central design decision in any dual-loop PFC controller, PID-based or otherwise, and it directly shapes the THD figure reported later in Section VIII.

Table II lists the design targets adopted for this work, drawn from typical single-phase appliance-class PFC ratings and from the IEC 61000-3-2 Class A harmonic limit against which the simulated result is checked.

IV. SYSTEM ARCHITECTURE

The power stage is the conventional single-switch boost converter: a bridge-rectified line voltage feeds the boost inductor L, a single MOSFET switch alternately stores energy in L and discharges it through the boost diode into the output capacitor C, which together with the load resistance R sets the 400 V DC bus. Because the focus of this work is the control loop rather than the switching cell itself, the power stage was implemented in Simulink as an averaged continuous-time model: the inductor and capacitor each appear as a single integrator block driven by their respective averaged voltage and current equations, with the switching duty ratio $d(t)$ entering through the $(1-d)$ term that couples the two integrators. This is a deliberate simplification, discussed further in Section X, that lets the control loops be designed and verified before committing to a switched, device-level model.

Control follows the average-current-mode structure common to commercial PFC controller ICs, arranged as two nested loops. The outer voltage loop compares the sensed output voltage against the 400 V reference and passes the error through a PID compensator (PID_Voltage) whose output sets the *amplitude* of the current the inner loop should track, not its instantaneous shape. That shape comes from a separate path: the sensed AC source voltage is scaled by a normalizing gain (Norm_Gain) to produce a unit-amplitude template of the rectified line waveform, and this template is multiplied by the voltage loop's output to produce the instantaneous current reference $i_{ref}(t)$. The inner current loop then compares $i_{ref}(t)$ against the sensed inductor current $i_L(t)$ and drives the error through a second PID compensator (PID_Current), whose output becomes the duty command applied to the averaged power-stage model. Figure 1 sketches this arrangement.

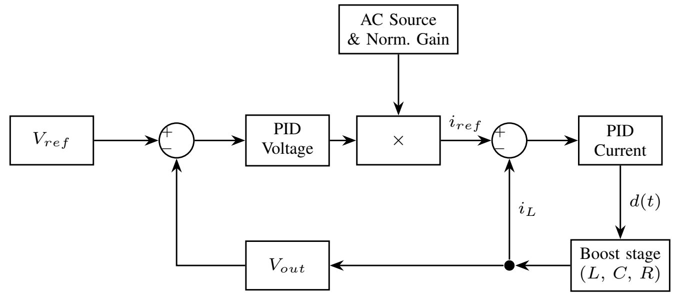

Fig. 1. Dual-loop average-current-mode control architecture: an outer voltage loop sets the amplitude of a rectified-sine current template, and an inner current loop forces the inductor current to follow it.

This arrangement deliberately separates two concerns that a single-loop controller would conflate: the voltage loop only ever needs to react to genuine changes in load or supply RMS value, while the current loop must react within a small fraction of the switching period to keep the inductor current tracking a continuously changing sinusoidal template. Splitting the bandwidth requirement this way is what allows a PID controller, despite its modest structure, to do a passable job on both fronts at once.

V. MATHEMATICAL MODELLING

A. Power Factor, Distortion and Harmonics

For a periodic, possibly non-sinusoidal current $i(t)$ drawn from a sinusoidal source $v(t)$, the real power, apparent power and reactive power are

$$P = V_{rms} I_{rms} \cos\varphi_1, \tag{1}$$

$$S = V_{rms} I_{rms}, \tag{2}$$

$$Q = V_{rms} I_{rms} \sin\varphi_1, \tag{3}$$

where φ_1 is the phase angle between the voltage and the *fundamental* component of the current. The power factor is then

$$PF = \frac{P}{S}. \tag{4}$$

It is convenient to split PF into a displacement term and a distortion term,

$$PF = \underbrace{\frac{I_1}{I_{rms}}}_{DF} \underbrace{\cos\varphi_1}_{DPF}, \tag{5}$$

where I_1 is the rms fundamental current and DF is the distortion factor. Total harmonic distortion is defined from the harmonic spectrum of the current as

$$THD = \frac{\sqrt{\sum_{h=2}^{\infty} I_h^2}}{I_1}, \tag{6}$$

and the distortion factor and THD are related by

$$DF = \frac{1}{\sqrt{1 + THD^2}}. \tag{7}$$

Equations (5) and (7) show why a controller that only corrects the displacement angle, such as a passive shunt capacitor, cannot fix the power factor of a rectifier load: the dominant error in that case is in DF, not DPF, and only an active current-shaping converter touches DF directly.

B. Averaged Boost Converter Model

With the switch and diode replaced by their averaged duty-dependent behaviour, the boost stage obeys

$$L\frac{di_L}{dt} = v_{in}(t) - (1 - d(t))\, v_{out}(t), \tag{8}$$

$$C\frac{dv_{out}}{dt} = (1 - d(t))\, i_L(t) - \frac{v_{out}(t)}{R}. \tag{9}$$

Equation (8) is the inductor volt-second balance: whenever the switch is on, the full line voltage appears across L; whenever it is off, the inductor sees the line voltage minus the (negated) output voltage. Equation (9) balances the current delivered to the capacitor by the diode against the current drawn by the load. These two equations, each implemented as a single integrator in the Simulink model, are the entire averaged plant the two PID loops must control.

C. Reference Generation and Error Signals

The instantaneous current reference is formed as

$$i_{ref}(t) = G_{norm}\, v_{in}(t)\, u_v(t), \tag{10}$$

where G_{norm} normalizes the sensed line voltage to unit amplitude and $u_v(t)$ is the (slowly varying) output of the voltage-loop PID compensator. The two error signals driving the controllers are

$$e_v(t) = V_{ref} - v_{out}(t), \tag{11}$$

$$e_i(t) = i_{ref}(t) - i_L(t). \tag{12}$$

VI. PID Controller Design

Each loop uses the standard PID structure with a filtered derivative term, as implemented in the Simulink continuous-time PID block:

$$C(s) = K_p + \frac{K_i}{s} + \frac{K_d N s}{s + N}, \tag{13}$$

where N limits the high-frequency gain of the derivative path so that switching noise on the sensed signals is not differentiated directly into the duty command. In both loops of this design $K_d = 0$, so (13) collapses to a pure PI compensator, $C(s) = K_p + K_i/s$, with N retained only as an inactive parameter of the controller block; this choice reflects a common practical concern that the inductor-current measurement in a real PFC stage carries enough switching ripple that a derivative term would amplify noise faster than it improves transient response.

The inner current loop must respond within a small fraction of the switching period to track a reference that itself changes shape continuously over the line cycle, so it was tuned for high bandwidth and a fast integral action: $K_p = 2.5$, $K_i = 500$. The outer voltage loop, by contrast, must stay slow enough that it does not chase the twice-line-frequency ripple riding on the sensed output voltage, since doing so would feed a 100 Hz component back into $i_{ref}(t)$ and create exactly the kind of low-order harmonic the converter is supposed to remove; it was tuned correspondingly gently: $K_p = 0.15$, $K_i = 8.0$. Both sets of gains were arrived at empirically, following the standard cascade-design guideline of separating the two loop bandwidths by roughly one decade and checking the resulting step response in simulation, rather than through a formal pole-placement or frequency-response procedure; this mirrors the practical tuning approach reported for Ziegler–Nichols-based PFC designs in [6] and motivates the metaheuristic alternatives discussed in Section IX.

TABLE III
Simulation Parameters

Parameter	Value
Input voltage	230 V rms, 50 Hz
Output voltage reference	400 V
Output capacitance, C	470 μF
Load resistance, R	200 Ω
Switching frequency, f_{sw}	100 kHz
Voltage-loop gains	$K_p = 0.15$, $K_i = 8.0$, $K_d = 0$, $N = 10$
Current-loop gains	$K_p = 2.5$, $K_i = 500$, $K_d = 0$, $N = 1000$
Simulation span	0–100 ms

For either loop, with plant transfer function $G(s)$ obtained by linearizing (8)–(9) about the operating point, the closed-loop response under unity feedback is

$$T(s) = \frac{C(s)\, G(s)}{1 + C(s)\, G(s)}. \tag{14}$$

Because $C(s)$ in this design is a PI compensator, the loop gain $C(s)G(s)$ contains a pole at the origin, which forces zero steady-state error to a step reference in both loops, consistent with the 0.062% output regulation error reported in Section VIII.

VII. Simulation Setup in MATLAB/Simulink

The complete model follows the architecture of Section IV almost block for block. A sine-wave source represents the 230 V, 50 Hz AC supply; a gain block (Norm_Gain) produces the unit-amplitude rectified template used to shape $i_{ref}(t)$; the voltage-loop summing junction (Sum_Volt) compares a constant reference block (Vout_Ref $= 400$) against the fed-back output voltage and drives the voltage PID block (PID_Voltage); a product block (Iref_Shape) multiplies the voltage-loop output by the normalized template; the current-loop summing junction (Sum_Curr) compares this product against the sensed inductor current and drives the current PID block (PID_Current); and the resulting duty signal is combined with constant and complementary-duty blocks (One_Const, One_Minus_D) before entering the averaged inductor and capacitor integrators (Sum_vL/L_inv and Sum_iC/C_inv), the latter loaded by a constant conductance block (G_load $= 1/200$) representing the 200 Ω resistive load. Table III lists the parameter values used.

Five scope groups (VoltageLoop, Main, CurrentLoop, PowerQuality and PIDOutputs) and a matching set of workspace logging signals (V_{out}, i_L, i_{ref}, V_s, the two loop errors, and the two PID outputs) were used to inspect transient behaviour, steady-state current shape, harmonic content and controller

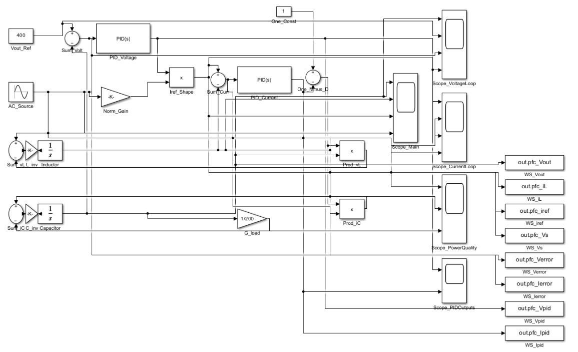

Fig. 2. Simulink implementation of the averaged boost PFC model, showing the dual-loop control architecture, the averaged plant dynamics, and the scope groups used for logging the simulation results.

TABLE IV
SIMULATED PERFORMANCE SUMMARY

Metric	**Value**
Mean output voltage	400.25 V
Output regulation error	0.062%
Output ripple (peak-to-peak)	14.32 V
True power factor	0.9412
Displacement power factor	≈0.99
Current THD	20.38%
Peak overshoot during start-up	≈17.5% (470 V peak)
Settling time (within ±2% of 400 V)	≈24 ms
IEC 61000-3-2 Class A (THD ≤ 8%)	Not met

saturation behaviour, the last of which is revisited in Section X.

VIII. RESULTS AND DISCUSSION

Table IV summarizes the steady-state and transient performance recorded from the simulation.

Figure 3 shows the key time-domain results from the averaged PFC model. The output voltage waveform rises quickly from start-up, overshoots the 400 V reference by roughly 17.5%, and settles within about 24 ms into a steady ripple band consistent with the 14.32 V peak-to-peak figure in Table IV. The same figure also shows the steady-state input voltage and inductor current relationship, where the current follows the line voltage closely except near the zero crossings.

Figure 4 makes the harmonic content and control-loop behaviour explicit. The dominant odd harmonics are visible in the current spectrum, the voltage and current loop errors settle quickly after start-up, and the PID duty outputs show the expected high initial transient before converging to their steady-state values.

Set against a typical uncompensated capacitor-input rectifier of comparable rating, which commonly presents a power factor in the 0.55–0.65 range and a current THD well above 100%, the dual-loop PID controller represents a substantial improvement: the displacement factor is pushed close to unity and the distortion factor, while not as high as it could be, still brings the true power factor above 0.94. The shortfall is specifically in the harmonic budget, and the 20.38% figure

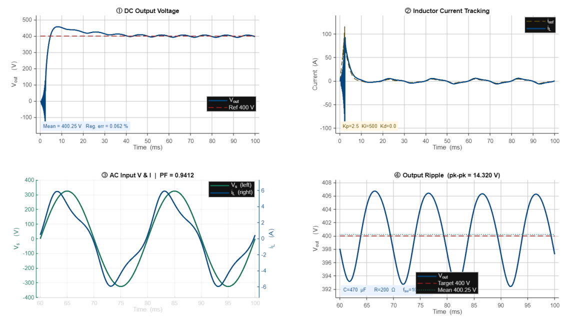

Fig. 3. Simulation waveform results from the averaged PFC model: DC output voltage, inductor current tracking, AC input voltage and output ripple around the 400 V reference.

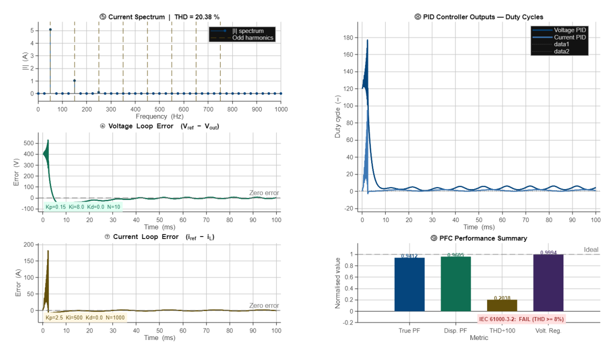

Fig. 4. Simulation analysis results: harmonic spectrum, voltage and current loop errors, PID outputs, and summary of the PFC performance metrics.

fails the IEC 61000-3-2 Class A limit of 8%, a result that is discussed further, alongside its likely causes, in Sections IX and X.

IX. COMPARISON WITH EXISTING METHODS

Table V places the result of this work alongside power-factor and THD figures reported for several of the control strategies reviewed in Section II, where such figures were explicitly stated in the source paper.

The pattern that emerges is unsurprising but worth stating plainly: every alternative in Table V that reports a harmonic figure reports one tighter than the 20.38% obtained here, and in every case the improvement comes from either a non-integer controller order, a metaheuristically searched gain set, or both. None of those alternatives, however, removes the need for a current loop and a voltage loop in roughly the architecture of Figure 1; they refine the compensator sitting inside that architecture rather than replacing it. That observation is exactly what keeps PID-based PFC relevant even as the literature moves toward more elaborate compensators: the dual-loop average-current-mode structure is largely architecture-agnostic, and a fixed-gain PID design such as the one presented here is a legitimate, low-risk starting point that can later be upgraded to a fractional-order or metaheuristically tuned compensator without redesigning the surrounding control structure, the sensing chain, or the power stage itself. For an industrial retrofit

TABLE V
COMPARISON WITH REPORTED PFC CONTROL RESULTS

Method	Source	PF	THD
Fixed-gain dual PID (this work)	[28]	0.9412	20.38%
Modulated-carrier control	[3]	>0.99	not reported
PSO-tuned fractional PI	[11]	≈0.99	lower than PI/fuzzy baseline
MPSO-tuned 4-param. PID	[10]	improved over manual	not directly stated
GWO-tuned FOPID	[13]	—	lower than integer PID
GEO-tuned fractional PI	[15]	high	1.85%

or a teaching laboratory where engineering time is scarcer than a few percentage points of harmonic margin, that upgrade path matters as much as the absolute number in Table V.

X. Advantages and Limitations

The dual-loop PID design earns its place primarily on the strength of its output regulation and its displacement power factor: a 0.062% steady-state voltage error and a displacement factor near unity are both achieved with a controller whose entire parameter set is four numbers per loop, easily implemented on inexpensive hardware and easily explained to anyone who has taken an introductory control course. The cascade structure itself, separating a fast current loop from a slow voltage loop, is sound and is the same structure used in essentially every PFC controller reviewed in Section II regardless of how its compensator is tuned.

Three limitations temper that result. First, and most visibly, the 20.38% THD fails the IEC 61000-3-2 Class A limit, and Section VIII traced that shortfall to current-shape flattening near the line-voltage zero crossings, a region where a fixed-gain PI controller has the least authority over the inductor current. Second, setting $K_d = 0$ in both loops, while a reasonable response to sensor noise, discards the one mechanism a PID controller has for anticipating fast transients, so the controller is, in effect, a PI controller wearing a PID label; a properly filtered derivative term, or a feedforward term based on the rate of change of the line voltage, is a natural next step rather than a structural change. Third, the simulation results in Table IV were obtained from the unsaturated PID output: the PIDOutputs scope showed the voltage-loop control signal transiently exceeding 100 during start-up, a value with no physical meaning once a duty cycle must lie between zero and one, which means the model as built omits the saturation and anti-windup logic that any real implementation would need and that could itself affect the reported overshoot and settling time once added.

XI. Future Scope

The comparison in Section IX points to several concrete upgrades that would close the gap between this design and the lowest-THD results reported in the recent literature, without discarding the underlying dual-loop architecture. Replacing the fixed PI gains with a metaheuristically searched set, following the particle-swarm or grey-wolf approaches in [9], [10], [13], is the most direct route and requires no change to the control structure itself. A fractional-order extension of either loop, in the spirit of [11], [13], [16], offers an additional tuning degree of freedom that several of the reviewed papers credit with a meaningfully lower THD, though it would need careful attention to real-time computability if implemented on a low-cost microcontroller rather than in simulation. Adding feedforward of the rectified line voltage's rate of change, to compensate for the loss of current-loop authority near the zero crossings identified in Section VIII, is a smaller change that targets the specific distortion mechanism observed here rather than the controller structure in general.

On the implementation side, the model used in this study is an averaged, continuous-time representation of the power stage; a natural next step is to rebuild the same control loops around a switched, device-level model (or a hardware-in-the-loop rig) to capture switching ripple, dead-time effects and the duty-cycle saturation and anti-windup behaviour flagged in Section X, none of which an averaged model can expose. Beyond that, extending the design to an interleaved or bridgeless power stage, following the topology comparisons in [1], [2], [8], would raise the achievable power level and efficiency without requiring a fundamentally different controller, and a physical prototype, along the lines of the one reported for the MPSO-tuned design in [10], would let the simulated figures in this paper be checked against measured hardware data.

XII. Conclusion

This paper has presented the design and simulated performance of a dual-loop, PID-controlled boost-type power factor correction stage built around an averaged Simulink model of a 230 V, 50 Hz, 800 W single-phase system. The outer voltage loop and inner current loop, both implemented as simple PI compensators, brought the output bus to within 0.062% of its 400 V reference and lifted the input power factor to 0.9412, while the resulting current THD of 20.38% fell short of the IEC 61000-3-2 Class A limit, a shortfall traced to current-shape flattening near the line-voltage zero crossings rather than to any gross controller instability. Placed against the metaheuristically tuned and fractional-order alternatives surveyed in Section II, the fixed-gain PID design trades a tighter harmonic budget for a controller that is simpler to specify, tune and maintain, and the comparison in Section IX suggests that this trade-off, rather than any inherent obsolescence, is what keeps classical PID control a live option in PFC design: the dual-loop architecture it sits inside is the same one every more elaborate compensator reviewed here also uses, which leaves a relatively low-risk path open from the design reported here toward the fractional-order, feedforward-augmented or metaheuristically tuned upgrades discussed in Section XI.

References

[1] J. R. Ortiz-Castrillón, G. E. Mejía-Ruíz, N. Muñoz-Galeano, J. M. López-Lezama, and S. D. Saldarriaga-Zuluaga, "PFC single-phase AC/DC boost converters: Bridge, semi-bridgeless, and bridgeless topologies," *Applied Sciences*, vol. 11, no. 16, p. 7651, 2021.

[2] Z. Chen, P. Davari, and H. Wang, "Single-phase bridgeless PFC topology derivation and performance benchmarking," *IEEE Trans. Power Electron.*, vol. 35, no. 9, pp. 9238–9250, Sep. 2020.

[3] J. Kim, H. Choi, and C. Won, "New modulated carrier controlled PFC boost converter," *IEEE Trans. Power Electron.*, vol. 33, no. 6, pp. 4772–4782, Jun. 2018.

[4] H. Espitia, "Review of converter circuits with power factor correction," *Technologies*, vol. 13, no. 6, p. 221, 2025.

[5] Z. Ortatepe and A. Karaarslan, "DSP-based comparison of PFC control techniques applied on bridgeless converter," *IET Power Electron.*, vol. 13, no. 2, pp. 317–323, Feb. 2020.

[6] N. F. Ibrahim, "Online adaptive power factor correction controller for DC-DC converters," *Int. J. Power Electron. Drive Syst.*, vol. 14, no. 1, 2023.

[7] C. H. K. Jensen, R. B. Lind, J. C. Hertel, A. M. Ammar, A. Knott, and M. A. E. Andersen, "A time-based control scheme for power factor correction boost converter," in *Proc. IEEE Nordic Circuits and Systems Conf. (NORCAS)*, 2019.

[8] N. Molavi, M. Maghsoudi, and H. Farzanehfard, "Quasi-resonant bridgeless PFC converter with low input current THD," *IEEE Trans. Power Electron.*, vol. 36, no. 7, pp. 7965–7972, Jul. 2021.

[9] J. P. Romero, D. Páez, B. Noriega *et al.*, "Bio-inspired PSO technique applied to PID sintonization for power factor correction in a boost converter," in *Proc. IEEE 5th Colombian Conf. Automatic Control (CCAC)*, 2021.

[10] J. G. Guarnizo, C. A. Torres-Pinzón, J. Bayona, D. Páez, J. P. Romero, B. Noriega, and L. Paredes-Madrid, "MPSO-based PID control design for power factor correction in an AC-DC boost converter," *Automatika*, vol. 64, no. 4, pp. 893–902, 2023.

[11] M. Demirtas and F. Ahmad, "Fractional fuzzy PI controller using particle swarm optimization to improve power factor by boost converter," *Int. J. Optim. Control Theor. Appl. (IJOCTA)*, vol. 13, no. 2, pp. 205–213, 2023.

[12] P. Kandasamy and K. Prem Kumar, "Efficient single-stage bridgeless AC to DC converter using grey wolf optimization," *Comput. Syst. Sci. Eng.*, vol. 43, no. 2, pp. 487–499, 2022.

[13] S. Lai and W. Wang, "Design of CCM boost converter using fractional-order PID controller optimized with gray wolf algorithm for power factor correction," *Int. J. Dyn. Control*, 2024.

[14] W. Liu, Y. Liu, and A. Wang, "Design of CCM boost converter using fractional-order PID and neural-network techniques for power factor correction," *J. Eng. Appl. Sci.*, vol. 71, no. 1, 2024.

[15] S. Vijayakumar and N. Sudhakar, "Golden eagle optimized fractional-order PI controller design for a PFC SEPIC converter in EV charging," *Sci. Rep.*, vol. 14, art. 20954, 2024.

[16] A. G. S. Sánchez, J. Soto-Vega, E. Tlelo-Cuautle, and M. A. Rodríguez-Licea, "Fractional-order approximation of PID controller for buck–boost converters," *Micromachines*, vol. 12, no. 6, p. 591, 2021.

[17] H. Mollaee, S. M. Ghamari, S. A. Saadat, and P. Wheeler, "A novel adaptive cascade controller design on a buck–boost DC–DC converter with a fractional-order PID voltage controller and a self-tuning regulator adaptive current controller," *IET Power Electron.*, vol. 14, no. 11, pp. 1920–1935, 2021.

[18] S. Saadat, S. M. Ghamari, and H. Mollaee, "Adaptive backstepping controller design on buck converter with a novel improved identification method," *IET Control Theory Appl.*, vol. 16, no. 5, pp. 1–14, 2022.

[19] S. M. Ghamari, F. Khavari, and H. Mollaee, "Lyapunov-based adaptive PID controller design for the buck converter," *Soft Comput.*, vol. 27, no. 9, pp. 5741–5750, 2023.

[20] V. Roosta, S. M. Ghamari, H. Mollaee, and M. Haddad Zarif, "A novel adaptive neuro linear quadratic regulator (ANLQR) controller design on DC-DC buck converter," *IET Renew. Power Gener.*, vol. 17, no. 5, pp. 87–92, 2023.

[21] F. Khavari, "Design of a novel robust type-2 fuzzy-based adaptive backstepping controller optimized with antlion algorithm for the buck converter," *IET Control Theory Appl.*, vol. 17, no. 9, pp. 758–762, 2023.

[22] C. González-Castaño, C. Restrepo, A. Sanz, A. Chub, and R. Giral, "DC voltage sensorless predictive control of a high-efficiency PFC single-phase rectifier based on the versatile buck-boost converter," *Sensors*, vol. 21, no. 15, p. 5107, 2021.

[23] S. Boddu and A. Ghosh, "Design of PFC converter with stand-alone inverter for microgrid applications," *J. Electr. Syst. Inf. Technol.*, vol. 10, no. 1, p. 36, 2023.

[24] K. Thirupura Sundari and M. G. Umamaheswari, "Predictive Riccati control for enhancing power quality using extended pq theory in wind energy-based conversion systems," *Sci. Rep.*, vol. 15, art. 34016, 2025.

[25] P. A. Tabbat, A. Khoshkbar-Sadigh, I. Talebian, V. Marzang, and E. Babaei, "Analysis of a high-efficient step-up converter with ZVS operation," in *Proc. 12th Power Electronics, Drive Systems and Technologies Conf. (PEDSTC)*, 2021, pp. 1–6.

[26] IEEE Std 519-2014, *IEEE Recommended Practice and Requirements for Harmonic Control in Electric Power Systems*, IEEE, 2014.

[27] International Electrotechnical Commission, *IEC 61000-3-2: Limits for Harmonic Current Emissions (Equipment Input Current $\leq$ 16 A per Phase)*, IEC, Geneva, Switzerland.

[28] K. Kesavaraj, *PID Control Design for Power Factor Correction*, GitHub repository, 2026. [Online]. Available: https://github.com/devkesav/PID-Control-Design-for-Power-Factor-Correction